

```

* =====.
* BI and Decision-MakingQuality in SMEs - Analysis Syntax.
* Dataset: DataSet1.csv (n = 186).
* Run each block in order. Output is the document of record.
* =====.

* Variable coding:
*   Job_Position 1=CEO 2=Senior Manager 3=Middle Manager
*                 4=Operational Manager 5=Other.
*   Industry      1=Manufacturing 2=Retail 3=Services
*                 4=Technology 5=Other.
*   All BI and DMQ items: 5-point Likert (1=Strongly Disagree ... 5=Strongly Agree).

* =====.
* STEP 0: Demographics (Table 1).
* =====.
FREQUENCIES VARIABLES=Job_Position Industry.

```

Frequencies

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Job_Position Industry.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Statistics

		Job_Position	Industry
N	Valid	186	186
	Missing	0	0

Frequency Table

Job_Position

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	13	7.0	7.0	7.0
	2	22	11.8	11.8	18.8
	3	37	19.9	19.9	38.7
	4	36	19.4	19.4	58.1
	5	78	41.9	41.9	100.0
	Total	186	100.0	100.0	

		Industry			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	14	7.5	7.5	7.5
	2	51	27.4	27.4	34.9
	3	53	28.5	28.5	63.4
	4	31	16.7	16.7	80.1
	5	37	19.9	19.9	100.0
	Total	186	100.0	100.0	

```

* =====.
* STEP 1: Reliability (Table 2 alphas).
* =====.
RELIABILITY VARIABLES=Strategic_BI_AvailabilityStrategic_BI_AlternativesStra
tegic_BI_Alignment
/SCALE('Strategic BI') ALL /MODEL=ALPHA.

```

Reliability

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=Strategic_BI _Availability Strategic_BI_Alternatives Strategic_BI_Alignment /SCALE('Strategic BI') ALL /MODEL=ALPHA.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Strategic BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded ^a	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.800	3

RELIABILITY VARIABLES=Tactical_BI_AvailabilityTactical_BI_Resource_AllocTactical_BI_Data_Decisions

/SCALE('Tactical BI') ALL /MODEL=ALPHA.

Reliability

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=Tactical_BI_Availability Tactical_BI_Resource_All oc Tactical_BI_Data_Decisions /SCALE('Tactical BI') ALL /MODEL=ALPHA.

Notes

Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Tactical BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded ^a	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.768	3

RELIABILITY VARIABLES=Operational_BI_AvailabilityOperational_BI_AgilityOperational_BI_Quality

/SCALE('Operational BI') ALL /MODEL=ALPHA.

Reliability

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality /SCALE('Operational BI') ALL /MODEL=ALPHA.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Operational BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded ^a	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.781	3

RELIABILITY VARIABLES=BI_Improves_DecisionsBI_Speed_AccuracyBI_Data_InformBI_Alternative_Solutions
/SCALE('DMQ') ALL /MODEL=ALPHA.

Reliability

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /SCALE('DMQ') ALL /MODEL=ALPHA.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: DMQ

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded ^a	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.845	4

```
RELIABILITY VARIABLES=BI_Training_LimitBI_Cost_BarrierBI_Data_Integration_Issue BI_Flexibility_Issue  
/SCALE('Challenges') ALL /MODEL=ALPHA.
```

Reliability

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /SCALE('Challenges') ALL /MODEL=ALPHA.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Challenges

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded ^a	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.650	4

* =====.

* STEP 2: Per-construct EFA (Table 2 loadings, KMO, Bartlett, %variance).

* Principal Axis Factoring, single factor. Read FACTOR MATRIX for loadings.

* Read EXTRACTION SUMS OF SQUARED LOADINGS for % variance.

* =====.

FACTOR VARIABLES=Strategic_BI_AvailabilityStrategic_BI_AlternativesStrategic_BI_Alignment

/PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.

Factor Analysis

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.

Notes

Syntax	FACTOR VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.	
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Maximum Memory Required	1860 (1.816K) bytes

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.694
Bartlett's Test of Sphericity	Approx. Chi-Square	177.873
	df	3
	Sig.	.000

Factor Matrix^a

	Factor 1
Strategic_BI_Availability	.708
Strategic_BI_Alternatives	.863
Strategic_BI_Alignment	.705

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 14 iterations required.

Communalities

	Extraction
Strategic_BI_Availability	.501
Strategic_BI_Alternatives	.745
Strategic_BI_Alignment	.497

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Extraction Sums of Squared Loadings

Factor	Total	% of Variance	Cumulative %
1	1.742	58.065	58.065

Extraction Method: Principal Axis Factoring.

FACTOR VARIABLES=Tactical_BI_AvailabilityTactical_BI_Resource_AllocTactical_BI_Data_Decisions
 /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.

Factor Analysis

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.

Notes

Syntax	FACTOR VARIABLES=Tactical_BI_Availability Tactical_BI_Resource_All oc Tactical_BI_Data_Decisions /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.	
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Maximum Memory Required	1860 (1.816K) bytes

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.689
Bartlett's Test of Sphericity	Approx. Chi-Square	146.395
	df	3
	Sig.	.000

Factor Matrix^a

	Factor 1
Tactical_BI_Availability	.665
Tactical_BI_Resource_Alloc	.810
Tactical_BI_Data_Decisions	.708

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 12 iterations required.

Communalities

	Extraction
Tactical_BI_Availability	.442
Tactical_BI_Resource_Alloc	.656
Tactical_BI_Data_Decisions	.502

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	1.600	53.333	53.333

Extraction Method: Principal Axis Factoring.

```
FACTOR VARIABLES=Operational_BI_AvailabilityOperational_BI_AgilityOperational_BI_Quality  
/PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.
```

Factor Analysis

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR VARIABLES=Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Maximum Memory Required	1860 (1.816K) bytes

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.702
Bartlett's Test of Sphericity	Approx. Chi-Square	155.402
	df	3
	Sig.	.000

Factor Matrix^a

	Factor 1
Operational_BI_Availability	.703
Operational_BI_Agility	.777
Operational_BI_Quality	.739

Extraction Method: Principal Axis
Factoring.

a. 1 factors extracted. 9 iterations required.

Communalities

	Extraction
Operational_BI_Availability	.494
Operational_BI_Agility	.604
Operational_BI_Quality	.546

Extraction Method: Principal Axis
Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	1.643	54.781	54.781

Extraction Method: Principal Axis Factoring.

FACTOR VARIABLES=BI_Improves_DecisionsBI_Speed_AccuracyBI_Data_InformBI_Alternative_Solutions

/PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.

Factor Analysis

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR VARIABLES=BI_Improves _Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Maximum Memory Required	2872 (2.805K) bytes

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.815
Bartlett's Test of Sphericity	Approx. Chi-Square	296.414
	df	6
	Sig.	.000

Factor Matrix^a

	Factor 1
BI_Improves_Decisions	.748
BI_Speed_Accuracy	.837
BI_Data_Inform	.759
BI_Alternative_Solutions	.698

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 7 iterations required.

Communalities

	Extraction
BI_Improves_Decisions	.560
BI_Speed_Accuracy	.700
BI_Data_Inform	.576
BI_Alternative_Solutions	.487

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	2.323	58.069	58.069

Extraction Method: Principal Axis Factoring.

* =====.

* STEP 3: Harman single-factor test (Section 3.4).

* All 13 items, one unrotated factor.

* Report FIRST FACTOR % from the SAME column you cite in 3.4 and 5.4.

* (Initial Eigenvalues column ~54%; Extraction Sums column ~51%.)

* =====.

FACTOR VARIABLES=Strategic_BI_AvailabilityStrategic_BI_AlternativesStrategic
_BI_Alignment

Tactical_BI_AvailabilityTactical_BI_Resource_AllocTactical_BI_Data_Decisions

Operational_BI_AvailabilityOperational_BI_AgilityOperational_BI_Quality
BI_Improves_DecisionsBI_Speed_AccuracyBI_Data_InformBI_Alternative_Solutions

/PRINT=EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.

Factor Analysis

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		FACTOR VARIABLES=Strategic_BI _Availability Strategic_BI_Alternatives Strategic_BI_Alignment Tactical_BI_Availability Tactical_BI_Resource_All oc Tactical_BI_Data_Decisio ns Operational_BI_Availabilit y Operational_BI_Agility Operational_BI_Quality BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /PRINT=EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.01
	Maximum Memory Required	21700 (21.191K) bytes

Factor Matrix^a

	Factor 1
Strategic_BI_Availability	.666
Strategic_BI_Alternatives	.765
Strategic_BI_Alignment	.709
Tactical_BI_Availability	.714
Tactical_BI_Resource_Alloc	.737
Tactical_BI_Data_Decisions	.675
Operational_BI_Availability	.678
Operational_BI_Agility	.703
Operational_BI_Quality	.698
BI_Improves_Decisions	.782
BI_Speed_Accuracy	.738
BI_Data_Inform	.719
BI_Alternative_Solutions	.656

Extraction Method: Principal Axis
Factoring.

a. 1 factors extracted. 4 iterations required.

Communalities

	Extraction
Strategic_BI_Availability	.443
Strategic_BI_Alternatives	.586
Strategic_BI_Alignment	.503
Tactical_BI_Availability	.510
Tactical_BI_Resource_Allo c	.543
Tactical_BI_Data_Decision s	.456
Operational_BI_Availability	.459
Operational_BI_Agility	.495
Operational_BI_Quality	.487
BI_Improves_Decisions	.612
BI_Speed_Accuracy	.544
BI_Data_Inform	.516
BI_Alternative_Solutions	.431

Extraction Method: Principal Axis
Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	6.586	50.659	50.659

Extraction Method: Principal Axis Factoring.

* =====.

* STEP 4: Descriptives + normality (Table 3 M/SD, skew, kurtosis).

* =====.

DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_
Making_Quality
/STATISTICS=MEAN STDDEV SKEWNESS KURTOSIS.

Descriptives

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality /STATISTICS=MEAN STDDEV SKEWNESS KURTOSIS.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Descriptive Statistics

	N Statistic	Mean Statistic	Std. Deviation Statistic	Skewness		Kurtosis Statistic
				Statistic	Std. Error	
Strategic_Mean	186	3.7330	.77329	-.725	.178	1.015
Tactical_Mean	186	3.7957	.70610	-.750	.178	1.943
Operational_Mean	186	3.8226	.74346	-.685	.178	.507
Decision_Making_Quality	186	3.8172	.74538	-.642	.178	.371
Valid N (listwise)	186					

Descriptive Statistics

	Kurtosis Std. Error
Strategic_Mean	.355
Tactical_Mean	.355
Operational_Mean	.355
Decision_Making_Quality	.355
Valid N (listwise)	

```
* =====.
* STEP 5: Spearman correlations (Table 3) on composite means.
* =====.
NONPAR CORR VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_M
aking_Quality
/PRINT=SPEARMAN TWOTAIL SIG.
```

Nonparametric Correlations

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		NONPAR CORR VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality /PRINT=SPEARMAN TWOTAIL SIG.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00
	Number of Cases Allowed	241979 cases ^a

a. Based on availability of workspace memory

Correlations

			Strategic_Mean	Tactical_Mean
Spearman's rho	Strategic_Mean	Correlation Coefficient	1.000	.694
		Sig. (2-tailed)	.	.000
		N	186	186
	Tactical_Mean	Correlation Coefficient	.694	1.000
		Sig. (2-tailed)	.000	.
		N	186	186
	Operational_Mean	Correlation Coefficient	.594	.658
		Sig. (2-tailed)	.000	.000
		N	186	186
	Decision_Making_Quality	Correlation Coefficient	.638	.635
		Sig. (2-tailed)	.000	.000
		N	186	186

Correlations

			Operational_Mean	Decision_Making_Quality
Spearman's rho	Strategic_Mean	Correlation Coefficient	.594	.638
		Sig. (2-tailed)	.000	.000
		N	186	186
	Tactical_Mean	Correlation Coefficient	.658	.635
		Sig. (2-tailed)	.000	.000
		N	186	186
	Operational_Mean	Correlation Coefficient	1.000	.677
		Sig. (2-tailed)	.	.000
		N	186	186
	Decision_Making_Quality	Correlation Coefficient	.677	1.000
		Sig. (2-tailed)	.000	.
		N	186	186

* =====.

* STEP 6: Main regression on FACTOR SCORES (Table 4, R², F, betas, CIs,

* VIF/tolerance, Durbin-Watson, Cook's, Mahalanobis).

* Cases must be in original file order for Durbin-Watson

* =====.

REGRESSION

```

/STATISTICS COEFF OUTS CI(95) R ANOVA COLLIN TOL
/DEPENDENT DMQ_Factor
/METHOD=ENTER Strategic_Factor Tactical_Factor Operational_Factor
/RESIDUALS DURBIN
/SAVE COOK MAHAL.

```

Regression

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS CI(95) R ANOVA COLLIN TOL /DEPENDENT DMQ_Factor /METHOD=ENTER Strategic_Factor Tactical_Factor Operational_Factor /RESIDUALS DURBIN /SAVE COOK MAHAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02

Notes

	Memory Required	2820 bytes
	Additional Memory Required for Residual Plots	0 bytes
Variables Created or Modified	MAH_4	Mahalanobis Distance
	COO_3	Cook's Distance

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Operational_Factor, Strategic_Factor, Tactical_Factor ^b	.	Enter

a. Dependent Variable: DMQ_Factor

b. All requested variables entered.

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.796 ^a	.634	.628	.60966442	2.255

a. Predictors: (Constant), Operational_Factor, Strategic_Factor, Tactical_Factor

b. Dependent Variable: DMQ_Factor

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	117.352	3	39.117	105.242	.000 ^b
	Residual	67.648	182	.372		
	Total	185.000	185			

a. Dependent Variable: DMQ_Factor

b. Predictors: (Constant), Operational_Factor, Strategic_Factor, Tactical_Factor

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-6.832E-17	.045		.000	1.000
	Strategic_Factor	.267	.071	.267	3.763	.000
	Tactical_Factor	.259	.075	.259	3.453	.001
	Operational_Factor	.364	.065	.364	5.573	.000

Coefficients^a

Model		95.0% Confidence Interval for B		Collinearity Statistics	
		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-.088	.088		
	Strategic_Factor	.127	.407	.399	2.503
	Tactical_Factor	.111	.406	.358	2.790
	Operational_Factor	.235	.493	.470	2.126

a. Dependent Variable: DMQ_Factor

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	(Constant)	Variance Proportions	
					Strategic_Factor	Tactical_Factor
1	1	2.408	1.000	.00	.06	.05
	2	1.000	1.552	1.00	.00	.00
	3	.351	2.618	.00	.35	.05
	4	.241	3.164	.00	.60	.90

Collinearity Diagnostics^a

Model	Dimension	Variance ...
		Operational_Factor
1	1	.06
	2	.00
	3	.86
	4	.08

a. Dependent Variable: DMQ_Factor

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	-3.0821428	1.4549369	.0000000	.79645258	186
Std. Predicted Value	-3.870	1.827	.000	1.000	186
Standard Error of Predicted Value	.046	.191	.084	.030	186
Adjusted Predicted Value	-3.1127129	1.4758117	-.0016912	.79631246	186
Residual	-2.03584790	1.83949029	.00000000	.60470099	186
Std. Residual	-3.339	3.017	.000	.992	186
Stud. Residual	-3.350	3.044	.001	1.004	186
Deleted Residual	-2.04852343	1.87293184	.00169123	.61941543	186
Stud. Deleted Residual	-3.448	3.116	.001	1.013	186
Mahal. Distance	.076	17.092	2.984	3.119	186
Cook's Distance	.000	.159	.006	.015	186
Centered Leverage Value	.000	.092	.016	.017	186

a. Dependent Variable: DMQ_Factor

* =====.

* STEP 7: Robustness check on standardized composite means (Section 4.3).

* =====.

REGRESSION

```

/STATISTICS COEFF OUTS R ANOVA
/DEPENDENT Decision_Making_Quality
/METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean

```

Regression

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS R ANOVA /DEPENDENT Decision_Making_Quality /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Memory Required	2852 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Operational_Mean, Strategic_Mean, Tactical_Mean ^b	.	Enter

a. Dependent Variable: Decision_Making_Quality

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.794 ^a	.630	.624	.45723

a. Predictors: (Constant), Operational_Mean, Strategic_Mean, Tactical_Mean

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	64.735	3	21.578	103.215	.000 ^b
	Residual	38.050	182	.209		
	Total	102.785	185			

a. Dependent Variable: Decision_Making_Quality

b. Predictors: (Constant), Operational_Mean, Strategic_Mean, Tactical_Mean

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.437	.196		2.227	.027
	Strategic_Mean	.254	.069	.264	3.702	.000
	Tactical_Mean	.278	.080	.263	3.486	.001
	Operational_Mean	.360	.066	.359	5.451	.000

a. Dependent Variable: Decision_Making_Quality

* =====.

* STEP 8: BI challenges (Table 5) M/SD and agreement %.

* Agreement % = proportion answering 4 or 5; read from frequency tables.

* =====.

DESCRIPTIVES VARIABLES=BI_Training_LimitBI_Cost_BarrierBI_Data_Integration_Issue BI_Flexibility_Issue
/STATISTICS=MEAN STDDEV.

Descriptives

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /STATISTICS=MEAN STDDEV.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Descriptive Statistics

	N	Mean	Std. Deviation
BI_Training_Limit	186	3.76	.963
BI_Cost_Barrier	186	3.52	1.041
BI_Data_Integration_Issue	186	3.48	.908
BI_Flexibility_Issue	186	3.33	.984
Valid N (listwise)	186		

FREQUENCIES VARIABLES=BI_Training_LimitBI_Cost_BarrierBI_Data_Integration_Issue BI_Flexibility_Issue

Frequencies

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Statistics

N		BI_Training_Limit	BI_Cost_Barrier	BI_Data_Integration_Issue	BI_Flexibility_Issue
	Valid	186	186	186	186
	Missing	0	0	0	0

Frequency Table

BI_Training_Limit

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	2	1.1	1.1	1.1
	2	18	9.7	9.7	10.8
	3	47	25.3	25.3	36.0
	4	74	39.8	39.8	75.8
	5	45	24.2	24.2	100.0
	Total	186	100.0	100.0	

BI_Cost_Barrier

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	8	4.3	4.3	4.3
	2	21	11.3	11.3	15.6
	3	55	29.6	29.6	45.2
	4	70	37.6	37.6	82.8
	5	32	17.2	17.2	100.0
	Total	186	100.0	100.0	

BI_Data_Integration_Issue

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	4	2.2	2.2	2.2
	2	18	9.7	9.7	11.8
	3	71	38.2	38.2	50.0
	4	70	37.6	37.6	87.6
	5	23	12.4	12.4	100.0
	Total	186	100.0	100.0	

BI_Flexibility_Issue

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	7	3.8	3.8	3.8
	2	26	14.0	14.0	17.7
	3	74	39.8	39.8	57.5
	4	57	30.6	30.6	88.2
	5	22	11.8	11.8	100.0
	Total	186	100.0	100.0	

* Cross-tabs by position and industry (Sections 4.4 and 5.3).

CROSSTABS TABLES=Job_Position BY BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue
/CELLS=ROW.

Crosstabs

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each table are based on all the cases with valid data in the specified range(s) for all variables in each table.

Notes

Syntax	CROSSTABS TABLES=Job_Position BY BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /CELLS=ROW.	
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Dimensions Requested	2
	Cells Available	349496

Case Processing Summary

	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Job_Position * BI_Training_Limit	186	100.0%	0	0.0%	186	100.0%
Job_Position * BI_Cost_Barrier	186	100.0%	0	0.0%	186	100.0%
Job_Position * BI_Data_Integration_Issue	186	100.0%	0	0.0%	186	100.0%
Job_Position * BI_Flexibility_Issue	186	100.0%	0	0.0%	186	100.0%

Job_Position * BI_Training_Limit Crosstabulation

% within Job_Position

		BI_Training_Limit					Total
		1	2	3	4	5	
Job_Position	1		15.4%	30.8%	7.7%	46.2%	100.0%
	2		4.5%	36.4%	40.9%	18.2%	100.0%
	3		5.4%	27.0%	51.4%	16.2%	100.0%
	4		8.3%	25.0%	50.0%	16.7%	100.0%
	5	2.6%	12.8%	20.5%	34.6%	29.5%	100.0%
Total		1.1%	9.7%	25.3%	39.8%	24.2%	100.0%

Job_Position * BI_Cost_Barrier Crosstabulation

% within Job_Position

		BI_Cost_Barrier					Total
		1	2	3	4	5	
Job_Position	1		7.7%	46.2%	23.1%	23.1%	100.0%
	2		13.6%	27.3%	50.0%	9.1%	100.0%
	3	8.1%	8.1%	24.3%	45.9%	13.5%	100.0%
	4		8.3%	27.8%	50.0%	13.9%	100.0%
	5	6.4%	14.1%	30.8%	26.9%	21.8%	100.0%
Total		4.3%	11.3%	29.6%	37.6%	17.2%	100.0%

Job_Position * BI_Data_Integration_Issue Crosstabulation

% within Job_Position

		BI_Data_Integration_Issue					Total
		1	2	3	4	5	
Job_Position	1	7.7%	7.7%	30.8%	38.5%	15.4%	100.0%
	2		9.1%	50.0%	36.4%	4.5%	100.0%
	3	2.7%	10.8%	37.8%	37.8%	10.8%	100.0%
	4		8.3%	30.6%	44.4%	16.7%	100.0%
	5	2.6%	10.3%	39.7%	34.6%	12.8%	100.0%
Total		2.2%	9.7%	38.2%	37.6%	12.4%	100.0%

Job_Position * BI_Flexibility_Issue Crosstabulation

% within Job_Position

		BI_Flexibility_Issue					Total
		1	2	3	4	5	
Job_Position	1	15.4%	23.1%	30.8%	15.4%	15.4%	100.0%
	2	4.5%	9.1%	36.4%	40.9%	9.1%	100.0%
	3	2.7%	16.2%	32.4%	37.8%	10.8%	100.0%
	4		8.3%	47.2%	30.6%	13.9%	100.0%
	5	3.8%	15.4%	42.3%	26.9%	11.5%	100.0%
Total		3.8%	14.0%	39.8%	30.6%	11.8%	100.0%

CROSSTABS TABLES=Industry BY BI_Training_LimitBI_Cost_BarrierBI_Data_Integration_Issue BI_Flexibility_Issue

/CELLS=ROW.

Crosstabs

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each table are based on all the cases with valid data in the specified range(s) for all variables in each table.
Syntax		CROSSTABS TABLES=Industry BY BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /CELLS=ROW.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Dimensions Requested	2
	Cells Available	349496

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Industry * BI_Training_Limit	186	100.0%	0	0.0%	186	100.0%
Industry * BI_Cost_Barrier	186	100.0%	0	0.0%	186	100.0%
Industry * BI_Data_Integration_Issue	186	100.0%	0	0.0%	186	100.0%
Industry * BI_Flexibility_Issue	186	100.0%	0	0.0%	186	100.0%

Industry * BI_Training_Limit Crosstabulation

% within Industry

		BI_Training_Limit					Total
		1	2	3	4	5	
Industry	1		7.1%	14.3%	42.9%	35.7%	100.0%
	2	2.0%	9.8%	35.3%	35.3%	17.6%	100.0%
	3	1.9%	7.5%	24.5%	49.1%	17.0%	100.0%
	4		9.7%	19.4%	38.7%	32.3%	100.0%
	5		13.5%	21.6%	32.4%	32.4%	100.0%
Total		1.1%	9.7%	25.3%	39.8%	24.2%	100.0%

Industry * BI_Cost_Barrier Crosstabulation

% within Industry

		BI_Cost_Barrier					Total
		1	2	3	4	5	
Industry	1	14.3%	7.1%	14.3%	42.9%	21.4%	100.0%
	2		13.7%	43.1%	35.3%	7.8%	100.0%
	3	5.7%	13.2%	24.5%	32.1%	24.5%	100.0%
	4	3.2%	6.5%	16.1%	51.6%	22.6%	100.0%
	5	5.4%	10.8%	35.1%	35.1%	13.5%	100.0%
Total		4.3%	11.3%	29.6%	37.6%	17.2%	100.0%

Industry * BI_Data_Integration_Issue Crosstabulation

% within Industry

		BI_Data_Integration_Issue					Total
		1	2	3	4	5	
Industry	1	7.1%	7.1%	7.1%	57.1%	21.4%	100.0%
	2	2.0%	9.8%	41.2%	33.3%	13.7%	100.0%
	3	1.9%	11.3%	43.4%	32.1%	11.3%	100.0%
	4		3.2%	32.3%	51.6%	12.9%	100.0%
	5	2.7%	13.5%	43.2%	32.4%	8.1%	100.0%
Total		2.2%	9.7%	38.2%	37.6%	12.4%	100.0%

Industry * BI_Flexibility_Issue Crosstabulation

% within Industry

		BI_Flexibility_Issue					Total
		1	2	3	4	5	
Industry	1		14.3%	35.7%	35.7%	14.3%	100.0%
	2	5.9%	11.8%	49.0%	23.5%	9.8%	100.0%
	3	1.9%	18.9%	41.5%	20.8%	17.0%	100.0%
	4	3.2%	9.7%	19.4%	54.8%	12.9%	100.0%
	5	5.4%	13.5%	43.2%	32.4%	5.4%	100.0%
Total		3.8%	14.0%	39.8%	30.6%	11.8%	100.0%

* End of syntax.

```
* =====.
* STEP 9: Sensitivity analysis - DMQ with the most BI-referential item.
* removed (Section 4.4, Robustness Check).
* Drops BI_Improves_Decisions(item references strategic/tactical/operational
* BI explicitly), re-derives a 3-item DMQ factor score, and re-runs the model.
* =====.
```

FACTOR

```
/VARIABLES BI_Speed_AccuracyBI_Data_InformBI_Alternative_Solutions
/MISSING LISTWISE
/ANALYSIS BI_Speed_AccuracyBI_Data_InformBI_Alternative_Solutions
/EXTRACTION PAF
/CRITERIA FACTORS(1)
```

```

/ROTATION NOROTATE
/SAVE REG(ALL DMQ3) .

```

Factor Analysis

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR /VARIABLES BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /MISSING LISTWISE /ANALYSIS BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /EXTRACTION PAF /CRITERIA FACTORS(1) /ROTATION NOROTATE /SAVE REG(ALL DMQ3).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Maximum Memory Required	2028 (1.980K) bytes
Variables Created	DMQ31	Regression factor score 1

Communalities

	Initial	Extraction
BI_Speed_Accuracy	.472	.663
BI_Data_Inform	.441	.589
BI_Alternative_Solutions	.391	.507

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.168	72.266	72.266	1.759	58.624	58.624
2	.462	15.404	87.669			
3	.370	12.331	100.000			

Extraction Method: Principal Axis Factoring.

Factor Matrix^a

	Factor 1
BI_Speed_Accuracy	.814
BI_Data_Inform	.767
BI_Alternative_Solutions	.712

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 9 iterations required.

* The saved score is named DMQ3_1. Sign is arbitrary; magnitudes match the paper.

REGRESSION

/STATISTICS COEFF OUTS CI(95) R ANOVA

/DEPENDENT DMQ3_1

/METHOD=ENTER Strategic_Factor Tactical_Factor Operational_Factor

Regression

Notes

Output Created		14-JUN-2026 21:14:06
Comments		
Input	Data	E:\Hoda's Desktop\ubis\Data\DataSet1.sav
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS CI(95) R ANOVA /DEPENDENT DMQ3_1 /METHOD=ENTER Strategic_Factor Tactical_Factor Operational_Factor.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.15

Warnings

Text: DMQ3_1 Command: REGRESSION

An undefined variable name, or a scratch or system variable was specified in a variable list which accepts only standard variables. Check spelling and verify the existence of this variable.

Execution of this command stops.

- * Expected (Section 4.4): $R^2 = .559$; Operational $\beta = .347$ ($p < .001$), .
- * Tactical = .252 ($p = .002$), Strategic = .236 ($p = .003$).

```
* =====.
* NOTE on CFA and HTMT (Section 4.2, Discriminant Validity).
* Base SPSS Statistics does not estimate confirmatory factor models. The
* four-factor vs one-factor CFA reported in the manuscript was estimated by
* maximum likelihood in a CFA-capable tool (e.g., AMOS, JASP, or R lavaan).
* HTMT is computed from the inter-item correlation matrix. Both are reproduced
* in the accompanying script cfa_htmt_sensitivity.py
* =====.

* End of syntax (v2).
```